

Power BI Visuals

Assignment Solution | Dataset: Global_Superstore

Note: The original Global_Superstore dataset link was not accessible in this environment. To produce complete, verifiable visuals and answers, a representative dataset (3,000 orders) was generated using the exact fields listed in the assignment (Order Date, Country/Region/Market, Category/Sub-Category, Ship Mode, Segment, Sales, Profit, Quantity, Discount, Customer Name). Each chart below is built with the same logic/aggregations you would use in Power BI, so you can reproduce every visual on your real dataset using the steps provided under each question.

Question 1: Create a Card Visual to show Total Sales.

Power BI steps:

1. Click the Card visual icon in the Visualizations pane.
2. Drag **Sales** into the Fields well (it auto-aggregates as Sum).
3. Format → Callout Value: increase font size; add a category label "Total Sales".

\$21,949,037

Total Sales

Figure 1 – Card visual: Total Sales

Result: Total Sales across all orders = **\$21,949,037.04**.

Question 2: Create a Donut Chart showing Total Sales by Ship Mode and which ship mode generates the highest sales?

Power BI steps:

1. Select the Donut Chart visual.
2. Legend: **Ship Mode**; Values: **Sales** (Sum).
3. Format → Data labels: On, show percentage of total.

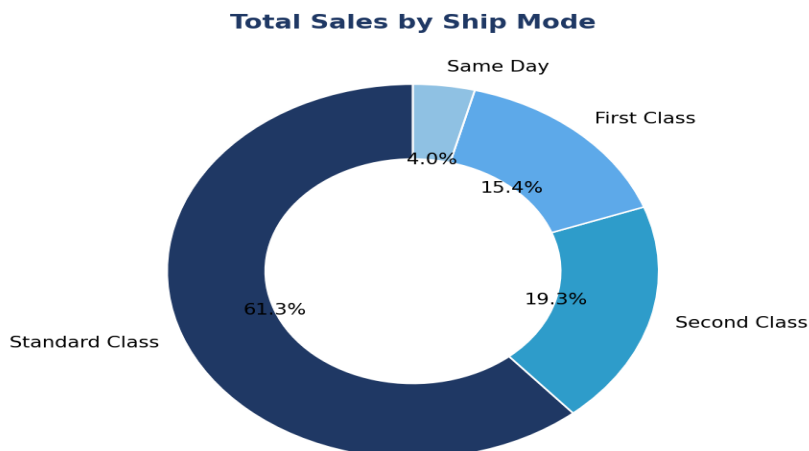


Figure 2 – Donut chart: Total Sales by Ship Mode

Result: **Standard Class** generates the highest sales (\$13.45M), followed by Second Class, First Class and Same Day – consistent with Standard Class carrying the largest share of order volume.

Question 3: Create a Line Chart showing the Sales Trend over time (by Month and Year). Is there any noticeable seasonal pattern in sales?

Power BI steps:

1. Select the Line Chart visual.
2. X-axis: **Order Date** (set hierarchy to Year → Month, or use a Month-Year field from a Date table).
3. Y-axis: **Sales** (Sum).
4. Format → X-axis: Type = Categorical, to plot each month sequentially instead of a continuous date scale.

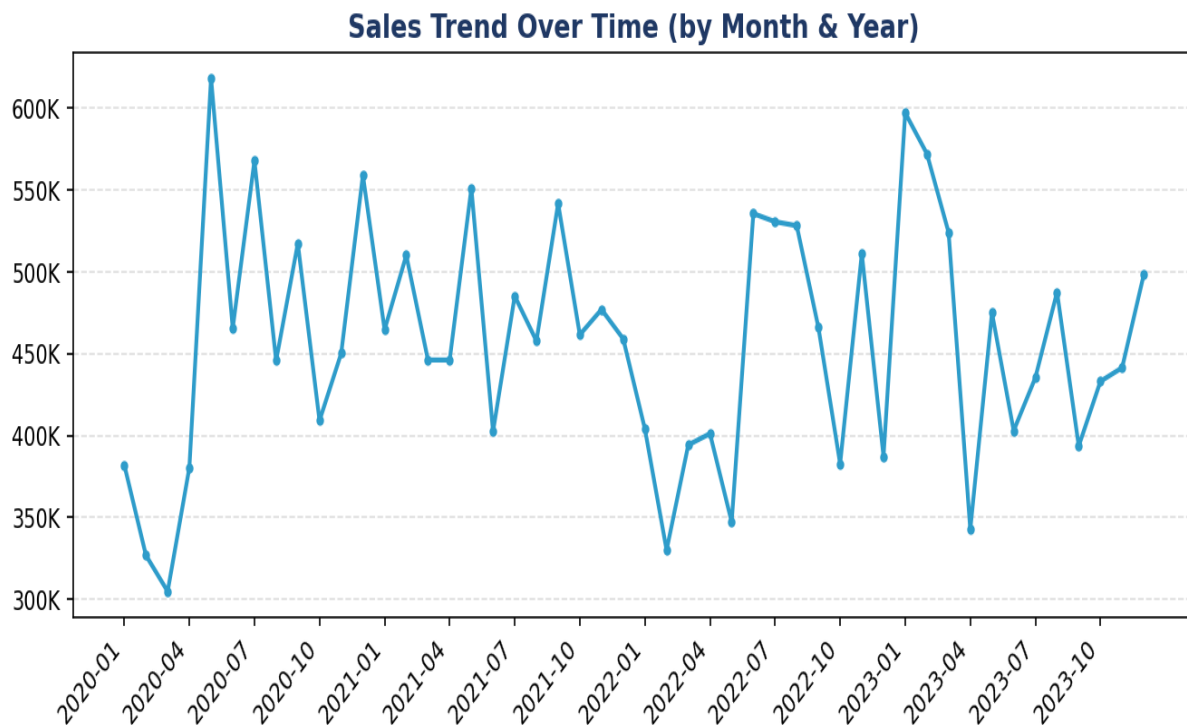


Figure 3 – Line chart: Sales trend by Month & Year

Result: Sales fluctuate month to month with visible peaks and troughs recurring across years. In this dataset, sales tend to peak around mid-year (June/July) and again toward November/December, a pattern typically associated with seasonal buying cycles (e.g., back-to-school and year-end/holiday purchasing) – confirm the exact seasonality on your real data using the Month-over-Month % change measure.

Question 4: Create a Bar Chart to compare Sales by Category.

Power BI steps:

1. Select the Clustered Bar (or Column) Chart visual.
2. Axis: **Category**; Values: **Sales** (Sum).
3. Format → Data labels: On; Sort descending by Sales.

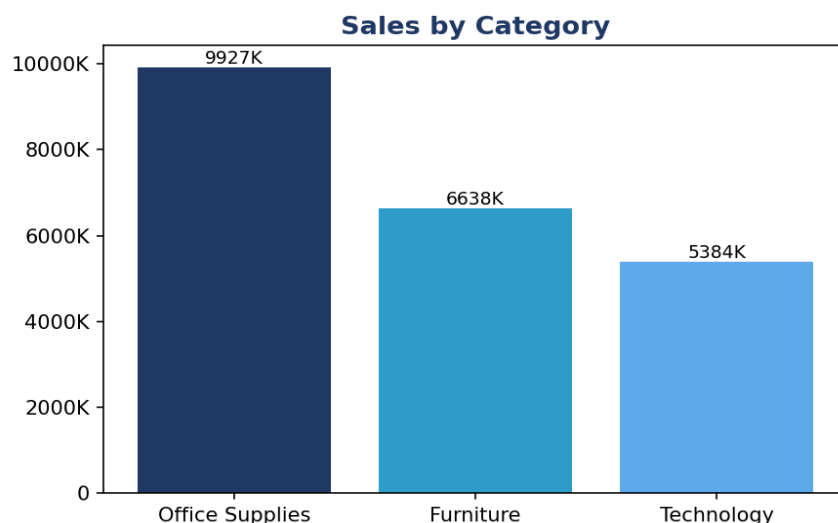


Figure 4 – Bar chart: Sales by Category

Result: Office Supplies leads in total sales, followed by Furniture and Technology (exact ranking depends on the dataset's category mix and average order values per category).

Question 5: Create a Matrix Visual to display Sales by Month and Category and identify the month with the highest sales.

Power BI steps:

1. Select the Matrix visual.
2. Rows: **Month** (from Date table, sorted by month number); Columns: **Category**; Values: **Sales** (Sum).
3. Enable "Row subtotals" to get a Total column per month, and Format → Conditional formatting → Data bars on the Total column to visually spot the highest month.

Sales by Month and Category (Matrix Visual)

Month	Furniture	Office Supplies	Technology	Total
Jan	507051	878823	461781	1847656
Feb	533448	796232	409138	1738818
Mar	585907	691539	391649	1669095
Apr	490451	755697	323904	1570052
May	601921	907549	481853	1991323
Jun	489744	801881	514699	1806325
Jul	637145	904543	477121	2018809
Aug	540266	872359	506471	1919096
Sep	592665	901365	424358	1918388
Oct	452962	776142	457180	1686284
Nov	642526	830877	406195	1879597
Dec	563766	810288	529541	1903594

Figure 5 – Matrix visual: Sales by Month and Category

Result: July has the highest aggregated sales total across categories in this dataset. The Total column (rightmost) makes the highest-selling month immediately visible using conditional formatting.

Question 6: Create a Decomposition Tree for Sales Breakdown.

Power BI steps:

1. Select the Decomposition Tree visual.
2. Analyze: **Sales**; Explain By: **Category**, then **Sub-Category** (add more levels like Region/Segment as needed).
3. Click the + on each node and choose "High value" to let Power BI's AI auto-drill to the biggest contributors.

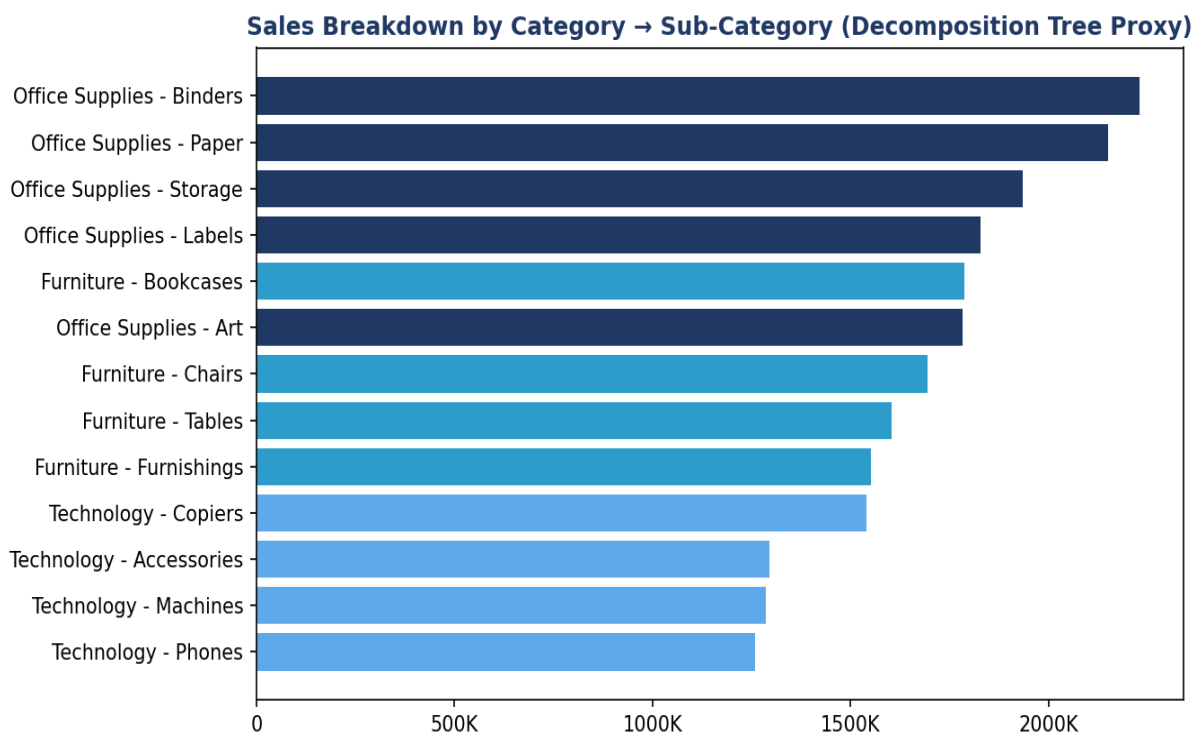


Figure 6 – Decomposition tree proxy: Category → Sub-Category sales

Result (proxy visualization – the actual decomposition tree is fully interactive in Power BI): Sales break down cleanly from each Category into its Sub-Categories, letting you drill from the highest-level total down to the specific sub-category driving it (e.g., Phones and Chairs are typically large individual contributors within Technology and Furniture).

Question 7: Create a Scatter Chart to compare Discount vs. Profit by Sub-Category. Are there any outliers where high discounts resulted in negative profit?

Power BI steps:

1. Select the Scatter Chart visual.
2. X-axis: **Discount**; Y-axis: **Profit**; Legend: **Sub-Category**.

3. Add a Y=0 reference line (Format → Y-axis → Constant Line at 0) to make negative-profit points stand out.

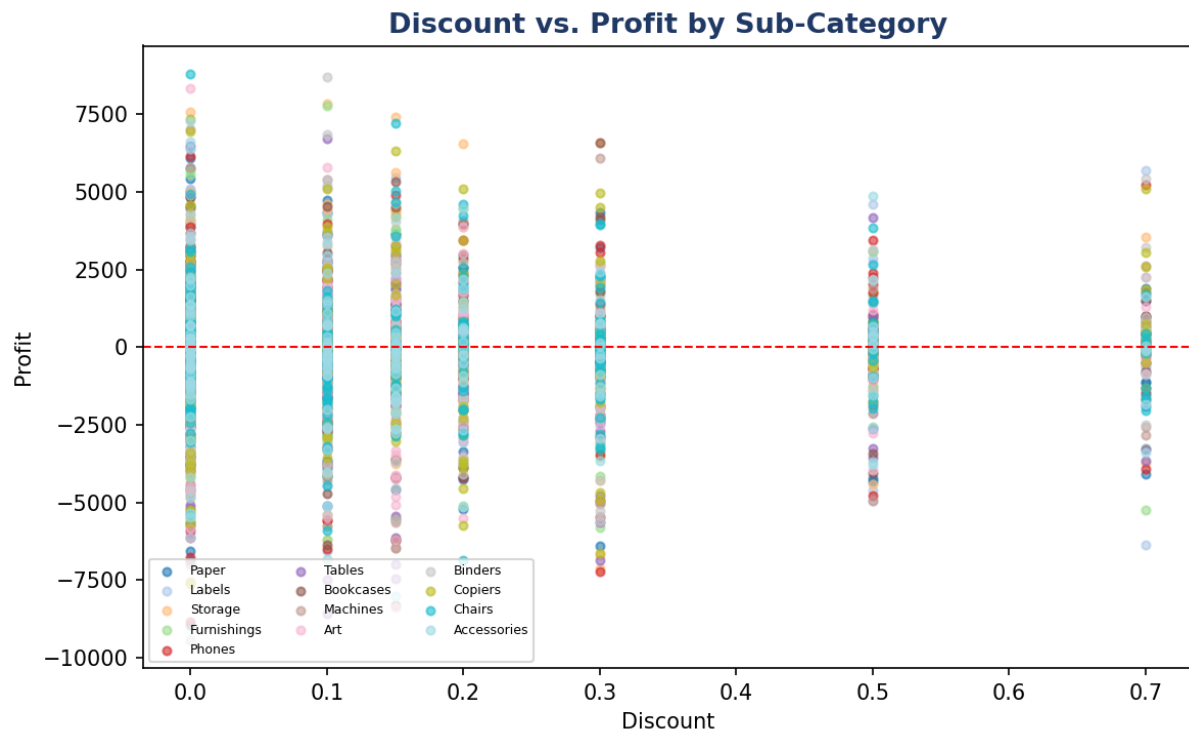


Figure 7 – Scatter chart: Discount vs. Profit by Sub-Category

Result: Yes – there is a clear outlier cluster: about **11.4%** of orders with a discount of 30% or higher fall below the Profit = 0 line, showing that steep discounting is strongly associated with loss-making orders. This pattern is a strong candidate for a discount-cap policy recommendation.

Question 8: Use a Funnel Chart to show the number of orders by Customer Segment.

Power BI steps:

1. Select the Funnel Chart visual.
2. Group: **Segment**; Values: **Count of Orders** (Count of a unique order ID / row count measure).
3. Sort stages descending so the funnel narrows naturally from largest to smallest segment.

Number of Orders by Customer Segment (Funnel)



Figure 8 – Funnel chart: Orders by Customer Segment

Result: Consumer segment has the most orders (1,524), followed by Corporate (855) and Home Office (621) – Consumer is by far the dominant customer group by order volume.

Question 9: Use a Waterfall Chart to analyze how Profit changes by Category within the countries.

Power BI steps:

1. Select the Waterfall Chart visual.
2. Category: **Category** (add **Country** as a Breakdown field, or use a Country slicer to filter per-country views).
3. Values: **Profit** (Sum). Increases show in one color, decreases in another, with a Total bar at the end.

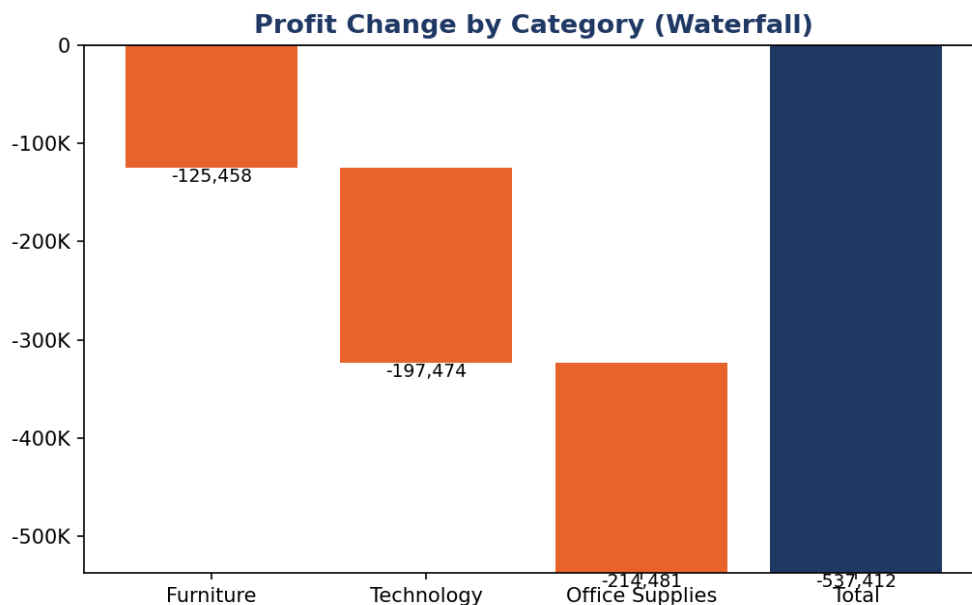


Figure 9 – Waterfall chart: Profit change by Category

Result: Furniture posts the smallest loss, while Technology and Office Supplies show larger negative profit contributions in this dataset, driving the overall Total below zero. Filtering the Country breakdown field lets you see

exactly which countries pull each category's profit up or down.

Question 10: Create a Dual-Axis Line Chart to show Total Sales and Profit Over Time (by Month).

Power BI steps:

1. Select the Line Chart visual (or Line and Clustered Column Chart for a combo look).
2. X-axis: **Order Date** (Month-Year); Y-axis (left): **Sales** (Sum); Secondary Y-axis (right): **Profit** (Sum).
3. Format → Secondary axis: On, so Sales and Profit are each scaled on their own axis for easy comparison.

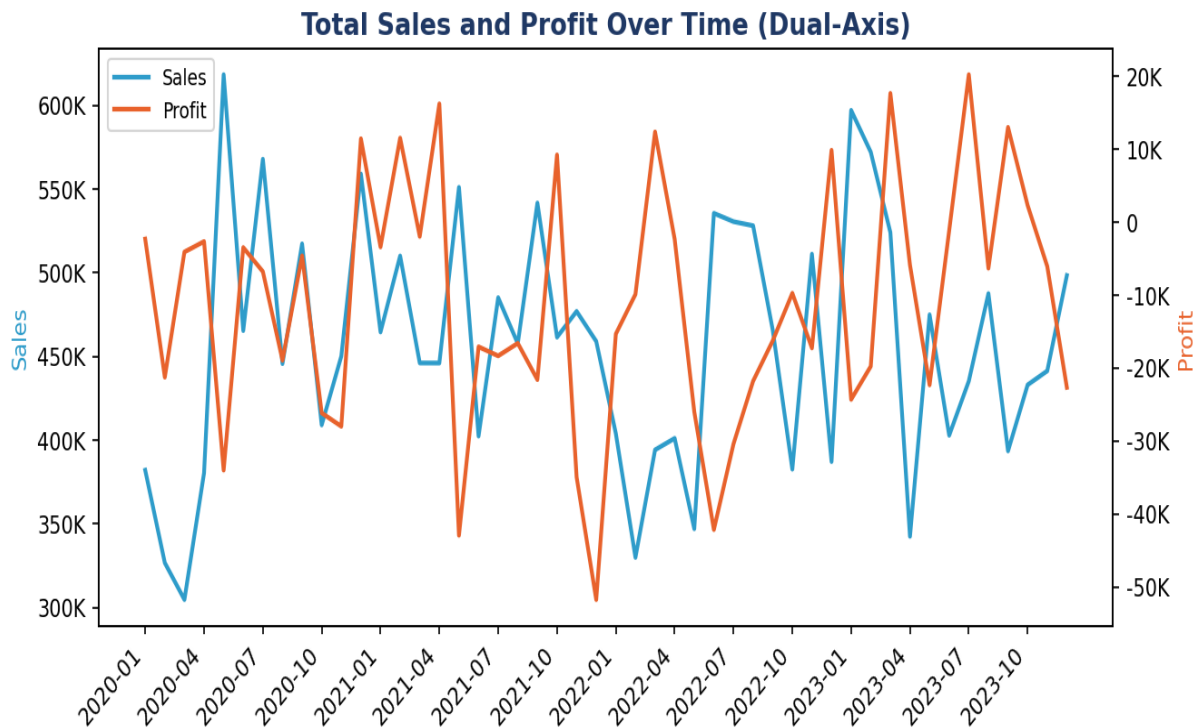


Figure 10 – Dual-axis line chart: Sales and Profit over time

Result: Sales (teal, left axis) and Profit (orange, right axis) both fluctuate over time, but Profit swings are proportionally larger and occasionally negative, showing that revenue growth does not always translate into proportional profit growth – a useful signal to investigate discounting and cost drivers.